

AE4 control of salivary transport: full system mathematics and parameter dependence

Tasks 44 and 45: standalone technical analysis

15 September 2026

Abstract

The frozen Task 40 system is analysed without removing intracellular or luminal carbon, alkalinity, volume, pH closure, electrical closure or AE4 regulation. Exact charge elimination leaves eleven independent dynamic coordinates. We prove conditional closure and balance results and distinguish them from local equilibrium calculations and finite time simulations. Equal cation routing cancels the explicit AE4 cycle in one stationary chloride balance; it does not cancel state mediated secretion sensitivity. The parent system has slow local modes of approximately 346 s in WT and 460 s in the AE4 null. A physiological NBC absent root exists at reduced AE4 loading, qualifying any universal necessity claim. Within the prescribed target selected Task 41 family, the first numerical crossing of a 30.3% cumulative deficit requires approximately 89.51% AE4 dependent stimulated CaCC recruitment. That family passes the inherited 600 s gates but later violates the pH gate. Local sensitivity identifies pump, NBC and NKCC capacities, CaCC recruitment and finite buffering as priorities for independent constraint. Unknown parameter ranges, thermodynamic tensions and experimental discrepancies prevent global robustness or mechanism validation claims.

1 Scope, checkpoint continuity and interpretation

This report continues the existing branch `analysis/task-44-full-system-mathematics`, recovered at `b7b775a0277d93263c4de339745806e5658ec9c2`. The complete numerical checkpoint independently replayed for this report is `f783785a863440df459e9c5530beba4c7eb59110`. Task 43 remains on the separate, unmerged parameter provenance branch; its final report checkpoint is `bc3e37645b9a7519dee0da9d1e2df55e291dd983`. The companion crosswalk reads numerical results at the exact Task 44 commit, with source file hashes. Neither a fit nor a new production model is introduced.

The archived `Marty.tex` fixes intracellular concentrations other than sodium, potassium and chloride, lumen concentrations, membrane potentials and cell volume. Those assumptions remove precisely the carbon, alkalinity, storage and osmotic responses at issue here. Its reduction is therefore not reused. The following scaling is an invertible coordinate change; every retained degree of freedom remains dynamic or is reconstructed by an exact constraint. The inherited algebraic electrical approximation is identified as an assumption, not justified retrospectively by a nonexistent capacitance parameter.

The report uses four levels of conclusion. A proved identity is conditional on its stated equations and domain. A local numerical result applies at the specified parameter point and branch of roots. A finite time numerical result applies to its stated stimulus, initial state and observation window. Unresolved questions include biological mechanism identity, global uniqueness, global stability, uncertainty ranges and robustness over those ranges.

2 The retained system and exact scaling

The analysis keeps the intracellular and luminal amounts of sodium, potassium, chloride, total inorganic carbon and total alkalinity; both volumes; and the AE4 regulatory fraction. Carbonate speciation and two membrane voltages are recovered algebraically at every evaluation. The older reduction in `Marty.tex` is therefore not used: neither a fixed pH nor a fixed volume can be inserted into the following identities. Alkalinity is a conserved acid–base coordinate, not bicarbonate alone.

Let N, H, E, B, J_4 denote inward NKCC1 cycles, NHE1 sodium uptake with proton extrusion, AE2 chloride uptake with bicarbonate extrusion, NBC cycles and AE4 cycles respectively. A positive NBC cycle imports one sodium and two bicarbonates. Write $P = P_a + P_b$ for the total pump cycles, K_a, K_b for potassium efflux from cell to lumen and bath, and C_a for chloride efflux from cell to lumen. The paracellular flux $j_{p,s}$ is positive from lumen to bath for species s . The carbon dioxide fluxes D_b, D_a are positive from bath and lumen into the cell. Water fluxes q_b, q_a, q_t, Q are positive from bath to cell, cell to lumen, bath to lumen, and lumen to sink respectively. These are signed fluxes, not their positive parts. The general AE4 source vector is

$$\nu = (\nu_{\text{Na}}, \nu_{\text{K}}, \nu_{\text{Cl}}, \nu_T, \nu_A), \quad S^4 = J_4 \nu.$$

Take $C_0 = [\text{Na}]_e$, $V_0 = V_i(0)$, $L_0 = V_l(0)$ and the initial stimulated WT NKCC1 *cycle* flux J_0 . Define

$$t_0 = \frac{C_0 V_0}{J_0}, \quad \tau = t/t_0, \quad \varepsilon = L_0/V_0, \quad v = V_i/V_0, \quad \ell = V_l/L_0, \quad x_s = \frac{n_{s,i}}{C_0 V_0}, \quad y_s = \frac{n_{s,l}}{C_0 L_0}.$$

Every ionic or carbon flux below is divided by J_0 , and every water flux by J_0/C_0 ; a hat marks these fluxes. Dimensionless concentrations are $c_{s,i} = x_s/v$ and $c_{s,l} = y_s/\ell$. Primes mean $d/d\tau$. The full dimensionless amount and volume equations are

$$\begin{aligned} x'_{\text{Na}} &= \hat{N} + \hat{H} + \hat{B} + \nu_{\text{Na}} \hat{J}_4 - 3\hat{P}, \\ x'_{\text{K}} &= \hat{N} + \nu_{\text{K}} \hat{J}_4 + 2\hat{P} - \hat{K}_a - \hat{K}_b, \\ x'_{\text{Cl}} &= 2\hat{N} + \hat{E} + \nu_{\text{Cl}} \hat{J}_4 - \hat{C}_a, \\ x'_T &= -\hat{E} + 2\hat{B} + \nu_T \hat{J}_4 + \hat{D}_b + \hat{D}_a, \\ x'_A &= \hat{H} - \hat{E} + 2\hat{B} + \nu_A \hat{J}_4, \\ v' &= \hat{q}_b - \hat{q}_a, \end{aligned} \tag{1}$$

$$\begin{aligned} \varepsilon y'_{\text{Na}} &= 3\hat{P}_a - \hat{j}_{p,\text{Na}} - \hat{Q} c_{\text{Na},l}, \\ \varepsilon y'_{\text{K}} &= -2\hat{P}_a + \hat{K}_a - \hat{j}_{p,\text{K}} - \hat{Q} c_{\text{K},l}, \\ \varepsilon y'_{\text{Cl}} &= \hat{C}_a - \hat{j}_{p,\text{Cl}} - \hat{Q} c_{\text{Cl},l}, \\ \varepsilon y'_T &= -\hat{j}_{p,\text{HCO}_3} - \hat{D}_a - \hat{Q} c_{T,l}, \\ \varepsilon y'_A &= -\hat{j}_{p,\text{HCO}_3} - \hat{Q} c_{A,l}, \\ \varepsilon \ell' &= \hat{q}_a + \hat{q}_t - \hat{Q}, \quad \theta_4 r' = \beta - r, \quad \theta_4 = 30/t_0. \end{aligned} \tag{2}$$

The neutral manifold is

$$x_A = x_{\text{Na}} + x_{\text{K}} - x_{\text{Cl}} - a_f, \quad y_A = y_{\text{Na}} + y_{\text{K}} - y_{\text{Cl}}, \quad a_f = \frac{n_{\text{fixed}}}{C_0 V_0}. \tag{3}$$

Eliminating these two amounts leaves eleven independent dynamic coordinates. This is exact: cell charge is conserved; luminal charge obeys $\dot{q}_l = -Q q_l / V_l$ and hence remains zero from zero initial charge. Alkalinity and its effect on pH remain in the reconstructed state.

2.1 Dimensionless constitutive laws

Here h_i and b_i denote the dimensionless hydrogen and bicarbonate concentrations from the carbonate closure below. All half concentrations and dissociation constants in this subsection are divided by C_0 ; numerical rate constants retain their declared unit conversions. For constant full stimulation the frozen NKCC1 and NHE1 multipliers are $m_N = 1.75$ and $m_H = 2.3$. More generally their separate clipped calcium coordinates have the frozen endpoints used by the production objects: the active NKCC1 endpoints are 0.058 and 0.1 μM , whereas NBC and NHE1 use 0.058 and 0.25 μM . This distinction matters away from full stimulation.

Put $X = c_{\text{Na},i}c_{\text{K},i}c_{\text{Cl},i}^2$. The Palk law actually selected in Task 40 [3] is

$$\hat{N} = e_N m_N \frac{\alpha_N A_1}{J_0 A_3} \frac{1 - \eta_2 X}{1 + \eta_4 X}, \quad \eta_2 = \frac{A_2 C_0^4}{A_1}, \quad \eta_4 = \frac{A_4 C_0^4}{A_3}. \quad (4)$$

The coefficients are $A_1 = 157.5$, $A_2 = 2.0096 \cdot 10^{-5}$, $A_3 = 1.0306$ and $A_4 = 1.3852 \cdot 10^{-6}$ in the repository's mM convention. The effective amount flux α_N comes from the frozen WT calibration, not an imported membrane density.

For NHE1, choose $k_* = 1000k_{1+}$, converting its printed rate per millisecond into a rate per second. Let barred rates be divided by k_* :

$$\begin{aligned} \bar{a} &= \frac{1000k_{1+}}{k_*} \frac{c_{\text{Na},e}}{c_{\text{Na},e} + K_{\text{Na},e}} \frac{K_{H,e}}{h_e + K_{H,e}}, & \bar{b} &= \frac{1000k_{2+}}{k_*} \frac{h_i}{h_i + K_{H,i}} \frac{K_{\text{Na},i}}{c_{\text{Na},i} + K_{\text{Na},i}}, \\ \bar{c} &= \frac{1000k_{1-}}{k_*} \frac{c_{\text{Na},i}}{c_{\text{Na},i} + K_{\text{Na},i}} \frac{K_{H,i}}{h_i + K_{H,i}}, & \bar{d} &= \frac{1000k_{2-}}{k_*} \frac{h_e}{h_e + K_{H,e}} \frac{K_{\text{Na},e}}{c_{\text{Na},e} + K_{\text{Na},e}}. \end{aligned}$$

Thus, with the exact printed Cha constants selected by production [2],

$$M_H = \frac{1}{1 + (1 + h_e/K_{M,e})(K_{M,i}/h_i)^3}, \quad \hat{H} = \frac{n_H k_*}{J_0} e_H m_H M_H \frac{\bar{a}\bar{b} - \bar{c}\bar{d}}{\bar{a} + \bar{b} + \bar{c} + \bar{d}}.$$

The printed $k_{2-} = 183 \text{ ms}^{-1}$ is retained: the small rounding discrepancy in microscopic reversibility is not silently repaired. The other acid–base flux laws are

$$\begin{aligned} \hat{E} &= e_2 \gamma_2 \tanh \left[\frac{\log(c_{\text{Cl},e} b_i / (c_{\text{Cl},i} b_e))}{w_2} \right], & \gamma_2 &= G_2 / J_0, \\ \hat{B} &= \gamma_B u \tanh \left[\frac{\log(c_{\text{Na},e} b_e^2 / (c_{\text{Na},i} b_i^2)) + \psi_b}{w_B} \right], & \gamma_B &= G_B / J_0, \\ \hat{D}_b &= \delta_b (c_{\text{CO}_2,e} - c_{\text{CO}_2,i}), & \hat{D}_a &= \delta_a (c_{\text{CO}_2,l} - c_{\text{CO}_2,i}). \end{aligned}$$

Here $\delta_{a,b} = P_{\text{CO}_2,a,b} C_0 / J_0$, $\psi_b = V_b / V_T$, $V_T = RT/F$, and u is the frozen NBC calcium coordinate. The tanh widths are dimensionless effective parameters.

For the inherited AE4 shared carrier, write k_4 for its common chloride attempt rate. Define

$$\begin{aligned} L_c &= \log(c_{\text{Cl},e} / c_{\text{Cl},i}), & L_s &= \log(c_{s,i} / c_{s,e}) + 2 \log(b_i / b_e), \quad s \in \{\text{Na}, \text{K}\}, \\ a_c^\pm &= \exp(\pm L_c / 2), & a_s^\pm &= (k_s / k_4) \exp(\pm L_s / 2), \\ U &= a_c^+ + a_{\text{Na}}^- + a_{\text{K}}^-, & W &= a_c^- + a_{\text{Na}}^+ + a_{\text{K}}^+, \quad \pi_i = U / (U + W), \quad \pi_o = W / (U + W). \end{aligned}$$

The scalar cycle is exactly

$$\hat{J}_4 = e_4 \gamma_4 (1 + 0.25r) \left[(a_{\text{Na}}^+ + a_{\text{K}}^+) \pi_i - (a_{\text{Na}}^- + a_{\text{K}}^-) \pi_o \right], \quad \gamma_4 = n_4 k_4 / J_0.$$

The frozen cooperative gate is absent. Task 40 applies equal ensemble sodium and potassium routing to this scalar cycle. The seven Catalán classes change its source vector but do not provide

new class specific kinetics. Local detailed balance of the inherited carrier edges does not by itself establish thermodynamic consistency of those substituted vectors.

To specify the electrical laws, scale each conductance G as $g = GV_T 10^{15} / (F J_0)$. The reversal potential for a valence z_s is $e_{s,uv} = z_s^{-1} \log(c_{s,v}/c_{s,u})$. With $g_{K,a}, g_{K,b}, g_{C,a}$ incorporating the frozen calcium gate a_{Ca} and background conductances, explicitly

$$g_{K,a} = g_{K,\text{total}} a_{Ca} f_K + g_{a,\text{bg}}, \quad g_{K,b} = g_{K,\text{total}} a_{Ca} (1 - f_K) + g_{b,\text{bg}}, \quad g_{C,a} = g_{Cl,a} a_{Ca},$$

the channel and paracellular fluxes are

$$\begin{aligned} \widehat{K}_a &= g_{K,a}(\psi_a - e_{K,il}), & \widehat{K}_b &= g_{K,b}(\psi_b - e_{K,ie}), \\ \widehat{C}_a &= -g_{C,a}(\psi_a - e_{Cl,il}), & \widehat{j}_{p,s} &= \frac{g_{p,s}}{z_s}(\psi_b - \psi_a - e_{s,le}). \end{aligned}$$

The apical pump is

$$\widehat{P}_a = \gamma_P f_P \frac{c_{Na,i}^3}{c_{Na,i}^3 + K_{P,Na}^3} \frac{c_{K,l}^2}{c_{K,l}^2 + K_{P,K}^2}, \quad \gamma_P = G_P / J_0,$$

and the basolateral expression replaces $f_P, c_{K,l}$ by $1 - f_P, c_{K,e}$. These pump rates are voltage independent in the frozen model. The calcium gate is $c_{Ca}^n / (c_{Ca}^n + K_{Ca}^n)$, using the same calcium units in numerator and denominator.

For water define dimensionless osmolarities

$$o_i = c_{Na,i} + c_{K,i} + c_{Cl,i} + c_{T,i} + (\beta_i + \iota_i)/v, \quad o_l = c_{Na,l} + c_{K,l} + c_{Cl,l} + c_{T,l},$$

where $\beta_i = n_{\text{buffer},i} / (C_0 V_0)$ and $\iota_i = n_{\text{other impermeant}} / (C_0 V_0)$. The bath expression includes its declared untracked osmolyte concentration. Exactly as in the production law, free hydrogen and hydroxide are omitted from osmolarity, and the tiny luminal buffer is not added as an osmotic particle. No new approximation is introduced by the present scaling. Then

$$\widehat{q}_b = \lambda_b(o_i - o_e), \quad \widehat{q}_a = \lambda_a(o_l - o_i), \quad \widehat{q}_t = \lambda_t(o_l - o_e), \quad \widehat{Q} = \omega \varepsilon (\ell - d_l)_+,$$

with $\lambda_j = L_j C_0^2 / J_0$, $\omega = k_{\text{out}} t_0$, and $d_l = V_{\text{dead}} / L_0$. These equations close (1)–(2) without fixing a volume, composition, carbon pool or voltage.

3 Carbonate closure and pH derivatives

Write p for pH, $h = 10^{-p}$ in molar units, $K_1 = 10^{-pK_1}$ and $K_2 = 10^{-pK_2}$. The carbonate fractions are

$$(\alpha_0, \alpha_1, \alpha_2) = \frac{(h^2, K_1 h, K_1 K_2)}{h^2 + K_1 h + K_1 K_2}, \quad f(p) = \alpha_1 + 2\alpha_2, \quad g(p) = \frac{1}{1 + 10^{pK_b - p}}.$$

In concentration units mM let $w(p) = 10^3(10^{p-pK_w} - 10^{-p}) = [\text{OH}] - [\text{H}]$. For either compartment, with fixed finite buffer amount n_b , the closure is the single equation

$$G(p; n_T, n_A, V) = n_T f(p) + n_b g(p) + V w(p) - n_A = 0. \quad (5)$$

Equivalently, for the cell, $x_A = x_T f(p) + \beta_i g(p) + v w(p) / C_0$; the lumen uses y_T, y_A, ℓ and $\beta_l = n_{b,l} / (C_0 L_0)$. Thus dilution of the finite buffer is retained exactly.

Proposition: unique speciation on the stated domain. Suppose $V > 0$, $n_T \geq 0$, $n_b \geq 0$, and all constants are finite and strictly positive in their concentration representation. For each real n_A , equation (5) has exactly one real pH. The solution depends smoothly on the retained amounts and volume on their open physical domain. For the production implementation one additionally requires $n_T > 0$ and

$$G(p_{\min}) \leq 0 \leq G(p_{\max});$$

smoothness within the executable domain is asserted with strict bracket inequalities. This numerical bracket is distinct from a physiological pH acceptance interval.

Proof. Let Z take the values 0, 1, 2 with probabilities $\alpha_0, \alpha_1, \alpha_2$. Differentiation gives

$$f'(p) = \log(10) \text{Var}(Z), \quad g'(p) = \log(10)g(1 - g), \quad w'(p) = \log(10)([\text{H}] + [\text{OH}]).$$

Consequently

$$G_p = \log(10)\{n_T \text{Var}(Z) + n_b g(1 - g) + V([\text{H}] + [\text{OH}])\} > 0. \quad (6)$$

The water term sends G to $-\infty$ as $p \rightarrow -\infty$ and to $+\infty$ as $p \rightarrow +\infty$. Continuity, strict monotonicity and the implicit function theorem prove the claims. The statement for the finite numerical bracket follows by monotonicity. $\square$

At fixed buffer amount the partial derivatives and trajectory derivative are

$$\frac{\partial p}{\partial n_A} = \frac{1}{G_p}, \quad \frac{\partial p}{\partial n_T} = -\frac{f}{G_p}, \quad \frac{\partial p}{\partial V} = -\frac{w}{G_p}, \quad \dot{p} = \frac{\dot{n}_A - f\dot{n}_T - w\dot{V}}{G_p}. \quad (7)$$

If the buffer amount is a parameter, $\partial p / \partial n_b = -g / G_p$. The very small explicit volume derivative at fixed amounts does not permit constant volume: volume changes all transported concentrations, osmolarity, electrical reversals and therefore $\dot{n}_T$ and $\dot{n}_A$. Nor does small free hydrogen concentration justify dropping carbon or buffering.

4 The coupled electrical closure

Use the dimensionless conductances and voltages defined above. Set $g_a = g_{K,a} + g_{C,a}$, $g_p = \sum_s g_{p,s}$, $c_a = -g_{K,a}e_{K,il} - g_{C,a}e_{Cl,il} + \hat{P}_a$ and $c_p = -\sum_s g_{p,s}e_{s,le}$. The apical equation gives

$$\psi_a(\psi_b) = \frac{g_p \psi_b - c_a + c_p}{g_a + g_p}.$$

The remaining current equation, including the electrogenic NBC, is

$$R(\psi_b) = g_{K,b}(\psi_b - e_{K,ie}) + \hat{P}_b + \hat{B}(\psi_b) + g_p(\psi_b - \psi_a(\psi_b)) + c_p = 0. \quad (8)$$

Proposition: unique electrical root. At any chemical state with positive concentrations and finite logarithms, suppose all conductances and NBC capacity are nonnegative, $g_a + g_p > 0$, and $s_0 = g_{K,b} + g_p g_a / (g_a + g_p) > 0$. If $w_B > 0$ and $u \in [0, 1]$, the mathematical current closure has a unique root on the real voltage line, smoothly dependent on the chemical state and parameters. The production routine further requires its physical voltage bracket $[-0.5, 0.5]$ V to contain that root.

Proof. Writing $L_B = \log(c_{\text{Na},e}b_e^2/(c_{\text{Na},i}b_i^2))$,

$$R'(\psi_b) = s_0 + \frac{\gamma B u}{w_B} \text{sech}^2((L_B + \psi_b)/w_B) > 0.$$

The NBC term is bounded whereas the passive contribution grows linearly with positive slope s_0 . Thus $R(\psi_b) \rightarrow \pm\infty$ at the two ends of the real line. Monotonicity and the implicit function theorem complete the argument. For any retained variable z , $\partial_z \psi_b = -R_z/R_{\psi_b}$; differentiating the expression for ψ_a gives its corresponding derivative. $\square$

On a domain with positive amounts and volumes, strict pH and voltage brackets and $V_l > V_{\text{dead}}$, these closures make the constant input chemical vector field smooth. At the outflow threshold it is locally Lipschitz but not differentiable. The regulatory law on $0 \leq r \leq 1$ has the smooth analytical extension $\dot{r} = (\beta - r)/30$ and capacity multiplier $1 + 0.25r$; its derivative at $r = 1$ means the derivative from the physical side, or equivalently that extension. The executable trial clamp outside $[0, 1]$ must not be differentiated by a centred difference across the boundary. Doing so gives a spurious regulator eigenvalue $-1/60$ instead of $-1/30 \text{ s}^{-1}$. These regularity statements prove local existence and uniqueness of trajectories in the stated domain. They do not prove global positivity, global equilibrium uniqueness or global stability.

5 Alkalinity balance and conditional capacity obstruction

Proposition: parent alkalinity budget. For the parent equal routing class C0, the intracellular balance is

$$\dot{n}_{A,i} = H + 2B - E - 2J_4. \quad (9)$$

Hence an intracellular stationary state must satisfy $2J_4 = H + 2B - E$. For $J_4 \neq 0$ define

$$\mathcal{A} = \frac{H + 2B - E}{2J_4} = 1 + \frac{\dot{n}_{A,i}}{2J_4}, \quad \mathcal{A}_0 = \frac{H - E}{2J_4}.$$

Here $\mathcal{A}_0$ removes the NBC contribution at the same state; it is not the result of resolving the NBC null system. For $J_4 = 0$ these ratios are undefined, and the relevant statement is the unnormalised balance $\dot{n}_{A,i} = H + 2B - E$.

Proof and finite time form. NHE1 proton extrusion adds one alkalinity equivalent, AE2 bicarbonate export removes one, NBC imports two, and the parent AE4 cycle removes two. Neutral carbon dioxide exchange, water transfer and the included potassium, sodium and chloride pathways have no direct alkalinity source. Integration yields, for every finite valid interval $[0, T]$,

$$2 \int_0^T J_4 dt = \int_0^T (H + 2B - E) dt + n_{A,i}(0) - n_{A,i}(T). \quad (10)$$

Changing volume is already retained because these are amount balances. Equation (7) then gives the pH change without identifying alkalinity change with pH change. $\square$

Corollary: a conditional obstruction without NBC. Suppose a declared admissible domain gives $H \leq H_{\text{max}}$ and $E \geq -E_{\text{max}}$. At an equilibrium with $B = 0$, $J_4 \leq (H_{\text{max}} + E_{\text{max}})/2$. More generally, maintaining a positive mean demand J_d for time T requires

$$2J_d \leq H_{\text{max}} + E_{\text{max}} + \frac{n_{A,i}(0) - n_{A,i}(T)}{T}$$

when NBC is absent. If $n_{A,i}(T) \geq A_{\text{min}}$, a demand greater than the capacity term can persist for at most $[n_{A,i}(0) - A_{\text{min}}]/[2J_d - H_{\text{max}} - E_{\text{max}}]$. This is conditional on the stated domain and demand, not a universal impossibility of secretion without NBC.

For the frozen fixed bath and NHE1 amount one can obtain a rigorous but conservative bound from the inherited pH condition $p_i \geq 6.6$. In the dimensional Cha notation a, d are fixed bath quantities, $b \leq b_0(h) = 1000k_{2+}h/(h + K_{H,i})$ and $c \geq 0$. Both $M_H(h)$ and $ab_0/(a + b_0 + d)$ increase with h . Therefore

$$H \leq n_H m_H M_H(h_{\max}) \frac{ab_0(h_{\max})}{a + b_0(h_{\max}) + d}, \quad h_{\max} = 10^3 10^{-6.6} \text{ mM}.$$

With $m_H \leq 2.3$ this yields $H_{\max} = 0.08202591 \text{ fmol/s}$; the tanh AE2 law gives $E_{\max} = 0.005 \text{ fmol/s}$. Consequently the corresponding NBC absent bound is $J_4 \leq 0.04351295 \text{ fmol/s}$. A *specified* demand of 0.075 fmol/s requires $B \geq 0.03148705 \text{ fmol/s}$ at equilibrium. The chosen demand is an illustrative model loading level, not a measured AE4 requirement. Neither this bound nor the source based kinetic law identifies the effective salivary carrier amounts.

6 Equal routing and general source projections

Proposition: exact chloride identity. For the full system with a constant AE4 source vector define

$$\kappa(\nu) = \nu_{\text{Cl}} - 2\nu_{\text{Na}} + \nu_A.$$

At every valid state,

$$C_a = 6P - H + \kappa(\nu)J_4 + 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i}. \quad (11)$$

For the parent equal routing vector $\kappa = 0$, yielding precisely

$$C_a = 6P - H + 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i}, \quad C_a^* = 6P^* - H^*.$$

Proof. From the three intracellular balances,

$$\begin{aligned} 2\dot{n}_{\text{Na},i} - \dot{n}_{A,i} - \dot{n}_{\text{Cl},i} &= 2(N + H + B + \nu_{\text{Na}}J_4 - 3P) \\ &\quad - (H - E + 2B + \nu_AJ_4) - (2N + E + \nu_{\text{Cl}}J_4 - C_a) \\ &= H - 6P - \kappa(\nu)J_4 + C_a. \end{aligned}$$

Rearrangement gives the claim. Neither carbon, volume nor pH was frozen. $\square$

Only the explicit AE4 term disappears from this stationary balance. AE4 still changes the chemical state, pump, NHE1, NKCC1, pH, volumes, membrane voltages and hence secretion. The proposition does not prove zero AE4 sensitivity, a small null phenotype for every admissible parameterisation, or observational equivalence of mechanisms. Away from equilibrium the three storage terms cannot be omitted. Integrating (11) gives the analogous finite time chloride budget by replacing each derivative integral with its endpoint amount difference.

The five relevant linear projections of a source vector are

$$\Pi_{\text{Cl}}\nu = \nu_{\text{Cl}}, \quad \Pi_T\nu = \nu_T, \quad \Pi_A\nu = \nu_A, \quad q\nu = \nu_{\text{Na}} + \nu_{\text{K}} - \nu_{\text{Cl}} - \nu_A, \quad \kappa\nu = -2\nu_{\text{Na}} + \nu_{\text{Cl}} + \nu_A.$$

The charge source is $(q\nu)J_4$. Thus a class with nonzero $q\nu$ would require its additional transported current in the electrical closure; merely changing amount sources would not preserve the neutral manifold. All seven frozen classes have $q\nu = 0$, so their explicit AE4 current is zero. On that neutral subspace, $\kappa\nu = \nu_{\text{K}} - \nu_{\text{Na}}$: the cancellation tests equal sodium and potassium source coefficients.

Class	ν_{Na}	ν_{K}	ν_{Cl}	ν_{T}	ν_{A}	κ
C0	$-1/2$	$-1/2$	1	-2	-2	0
C1	1	-1	1	-1	-1	-2
C2	1	-2	1	-1	-2	-3
C3a	0	-1	1	-2	-2	-1
C3b	-1	0	1	-2	-2	1
C4a	0	-1	1	-1	-2	-1
C4b	-1	0	1	-1	-2	1

Each individual balance annihilates the kernel of its displayed linear functional. For example a change in ν_{T} alone is invisible to $\Pi_{\text{Cl}}, \Pi_{\text{A}}, q, \kappa$ but not to carbon balance. This is exactly why C3a/C4a and C3b/C4b have the same listed stationary chloride coefficient but different carbon signatures. Jointly the five projections have rank five: if chloride, carbon and alkalinity projections vanish, $q = 0$ gives $\nu_{\text{K}} = -\nu_{\text{Na}}$, and $\kappa = 0$ then gives $\nu_{\text{Na}} = \nu_{\text{K}} = 0$. No nonzero direction is lost by all five together. Equality in one projected balance is not equality of molecular mechanisms or of the whole dynamical response.

6.1 Independent verification and scaling implications

The analysis script `verify_exact.py` independently assembles carbonate chemistry, all selected kinetic laws, the nonlinear electrical solve, finite lumen transport, changing volumes and the thirteen amount, volume and regulatory derivatives. It compares these with production at 84 random valid states spanning all seven source classes and varied AE4 expression. It also verifies the pH derivative formula at two perturbation sizes, the positive electrical derivative, the exact charge and chloride identities, and the correct inward regulatory derivative at $r = 1$. Rational source coefficients are retained in `output/exact_verification.json`. These implementation checks support the algebra but do not turn random samples into a global theorem.

Numerically $\varepsilon = 0.072515$ and $\theta_4 = 0.031977$; $\gamma_B = 0.515137$, $\gamma_2 = 0.022260$ and $\gamma_4 = 0.668363$. The water coefficients are $\lambda_a = 404.37$, $\lambda_b = 4820.57$ and $\lambda_t = 24.34$; $\omega = 187.64$. These suggest some rapidly adjusting variables but are insufficient to justify freezing either volume: their driving osmotic differences and coupling to amounts matter. The verified dimensional recovery retains them all. The small dissociation concentrations, luminal buffer amount and inherited carrier relaxation ratio are recorded separately in the machine readable groups. A small buffer amount is a numerical convention here, not a measured physiological asymptotic limit. There is no membrane capacitance or differential voltage equation in the frozen system, so no electrical relaxation parameter can be inferred from its conductance groups. The algebraic voltage law is an inherited modelling assumption, not a new singular perturbation theorem proved by this analysis.

7 Numerical scales and their interpretation

Quantity	Value	Units
C_0	145	mM
V_0	1.45333439	pL
L_0	0.10538841	pL
J_0	0.22461808	fmol/s
t_0	938.18578	s
ϵ_L	0.072515	1
θ_4	0.031977	1
G_B/J_0	0.515137	1
G_2/J_0	0.022260	1
$L_a C_0^2/J_0$	404.366	1
$L_b C_0^2/J_0$	4820.571	1
$L_t C_0^2/J_0$	24.337	1

Here the table's ϵ_L is the ϵ in the equations. The induced cell and lumen amount scales are 210.733487 and 15.2813191 fmol; the induced water flux scale is $J_0/C_0 = 0.00154909$ pL/s. For dimensional recovery multiply cell amounts by $C_0 V_0$, lumen amounts by $C_0 L_0$, the two volumes by their distinct scales, time by t_0 , ionic fluxes by J_0 and water fluxes by J_0/C_0 . pH is already dimensionless. The thermal voltage is 26.726659 mV.

Group	Meaning	Value
$n_{b,i}/(C_0 V_0)$	Cell buffer amount	0.14236
$n_{b,l}/(C_0 L_0)$	Lumen buffer amount	6.54394e-11
$n_{\text{fixed}}/(C_0 V_0)$	Fixed cell charge	0.370429
$n_{\text{other}}/(C_0 V_0)$	Other cell osmoles	0.59174
K_1/C_0	First carbonate dissociation	5.47813e-06
K_2/C_0	Second carbonate dissociation	3.45646e-10
K_w/C_0^2	Water dissociation	4.75624e-13
$K_{b,i}/C_0$	Cell buffer dissociation	6.89655e-07
$K_{b,l}/C_0$	Lumen buffer dissociation	6.89655e-07
$P_{\text{CO}_2,b} C_0/J_0$	Basolateral carbon dioxide exchange	12.9108
$P_{\text{CO}_2,a} C_0/J_0$	Apical carbon dioxide exchange	6.4554
$J_4(0^+)/J_0$	Initial AE4 to NKCC cycle ratio	0.0219659
$J_4(0^+)/(2J_0)$	Initial chloride loading ratio	0.0109829
t_{carrier}/t_0	Inherited carrier relaxation ratio	0.000139211

The carbonate constants in this table use mM before division by C_0 ; K_w uses mM squared. These units differ from the molar representation used to define pH in the proof, with the conversion shown there. The small initial AE4 loading ratio is evaluated at the inherited WT state under immediate full stimulation. It is not a sustained loading fraction. The reference carrier capacity ratio is $\gamma_4 = 0.668363$, the pump capacity ratio is 0.356160 and the outflow group is 187.637155. The Palk concentration groups are $\eta_2 = 56.402853$ and $\eta_4 = 594.147609$. Dimensionless maximal channel conductances are 38.722939 for CaCC and 17.265005 for potassium; the paracellular sodium, potassium, chloride and bicarbonate values are respectively 0.369964, 0.061661, 0.308304 and 0.061661. Stimulus gates multiply the channel capacities as specified in the constitutive laws.

Small dissociation concentrations do not imply a dispensable alkalinity coordinate. The cell buffer amount is appreciable relative to the amount scale. The small lumen buffer is an inherited numerical convention, whereas finite lumen carbon remains dynamical. Large water conductances allow small osmotic imbalances to move appreciable water. No uniform error bound or normally

attracting limiting manifold has been established that would permit freezing either volume or lumen composition. The present results therefore make no singular perturbation reduction.

8 Verification, equilibrium framework and compensation

8.1 Production identity and dimensional recovery

The analysis loads the actual frozen Task 40 objects and WT conserved initial state. It checks all six stimulated parameter-object hashes and the resting state hash against the frozen manifest, rather than rebuilding from dataclass defaults. The source audit compares 150 entries with the Task 40 parent manifest: 149 match; the sole difference is `AGENTS.md`, an instructional metadata change. All listed production Python sources match their recorded Git blobs. The recovery audit also found no differences in the protected scientific tree relative to the main head it inspected. The scientific base is `c4d4f207f702d8eee09ab94d2d90ae90552dd641`; the recovered Task 44 checkpoint is `b7b775a0277d93263c4de339745806e5658ec9c2`. The complete audit and object hashes are retained in `output/source_audit.json`. For an unambiguous initial-state anchor, the frozen thirteen-state SHA256 is

`a0b96ef16e5967de4dfe2bdabb882f2a28d478e211f920bc19cd735a5a143acb`

Dimensional recovery uses $n_{s,i} = C_0 V_0 x_s$, $n_{s,l} = C_0 L_0 y_s$, $V_i = V_0 v$, $V_l = L_0 \ell$ and $t = t_0 \tau$. If $\hat{I}$ is the integrated dimensionless water observable scaled by V_0 , then $I = V_0 \hat{I}$ and $Q = (J_0/C_0) \hat{Q}$. Separate dimensional and dimensionless integrations through 600 s differ by at most 7.524×10^{-10} in a state component divided by its declared scale, over the saved comparison times. A separate check recovers cumulative water to 1.443×10^{-15} pL. The reproduced original sampled WT output is 0.992524545 pL, compared with frozen 0.992524544 pL; its adaptive integral is 0.992544443 pL. These two observables have different numerical definitions. The WT, 5% expression and null reference checks reproduce all 34 diagnostic columns at eleven frozen display times, with a largest cumulative-output discrepancy of 6.74×10^{-10} pL. These are numerical equivalence checks on the retained equations, not a reduced-model fit.

8.2 Executable domain and inherited acceptance gates

The carbonate executable bracket is $[3, 11]$ for pH in both compartments. For the smooth local analysis, require $G(3) < 0 < G(11)$ for the amount residual in (5), positive transported concentrations and positive volumes. The NBC electrical solver brackets the basolateral voltage on $[-0.5, 0.5]$ V; strict bracketing of its residual defines the interior domain used for differentiation. The apical voltage is then recovered from the coupled current equation, without imposing a separate apical bracket. The numerical solvers can accept an exact endpoint root; that fact does not extend the stated smooth interior domain. Require $V_l > V_{\text{dead}}$ to remain on the differentiable, active-outflow branch. The physical regulatory range is $0 \leq r \leq 1$; its boundary derivative uses the inward rule or smooth extension already specified above.

The inherited physiological checks are more restrictive than numerical speciation. They require all twelve amount/volume components to be strictly positive and all states, derivatives and diagnostics to be finite, together with the following intracellular bounds:

Quantity	Accepted domain
Sodium	$[\text{Na}]_i < 40 \text{ mM}$
Potassium	$50 \leq [\text{K}]_i \leq 200 \text{ mM}$
Chloride	$30 \leq [\text{Cl}]_i \leq 80 \text{ mM}$
pH	$6.6 \leq \text{pH}_i \leq 7.3$
Bicarbonate	$0 < [\text{HCO}_3]_i < 100 \text{ mM}$
Total inorganic carbon	$[T]_i > 0$
Cell volume	$0 < V_i < 3 \text{ pL}$

The same diagnostics check charge, carbon and buffer accounting to 10^{-10} fmol/s, water accounting to 10^{-12} pL/s, each membrane current residual to 10^{-20} A, and compartmental speciation alkalinity to 10^{-9} mM. The declared diagnostic-key set must match, and NBC's signed charge and carbon sources must equal $-B$ and $2B$ exactly in the diagnostic representation. These last two sources are not zero residuals. No additional physiological luminal pH interval has been introduced.

The supplemental full-gate verification independently re-solves 128 parameter-perturbed WT/null roots and 256 nearby expression roots used for compensation validation. All 384 pass these inherited checks; the largest conservation residual divided by its own tolerance is 4.086×10^{-5} . The largest dimensionless independent equilibrium residual is 6.880×10^{-12} . These results are retained in `output/sensitivity_full_gate_verification.json`; gate acceptance does not remove the separate thermodynamic qualifications.

8.3 An eleven-coordinate implicit-function problem

Let

$$z = (x_{\text{Na}}, x_{\text{K}}, x_{\text{Cl}}, x_T, v, y_{\text{Na}}, y_{\text{K}}, y_{\text{Cl}}, y_T, \ell, r) \in \mathbb{R}^{11}, \quad \frac{dz}{d\tau} = F(z, \mu).$$

The two alkalinity coordinates are reconstructed by (3); both carbon amounts, both volumes and the regulatory state remain dynamic. The parameter μ denotes expression or one specified active model setting, with the scaling fixed at the reference values. Carbonate and voltage closures are evaluated inside F . Constant full stimulation defines the equilibrium problem $F(z^*, \mu) = 0$. The numerical solver finds ten positive chemical/volume coordinates in logarithmic variables and sets the *equilibrium* condition $r^* = 1$. All eleven equations are subsequently checked, and all eleven coordinates are retained in the dynamic Jacobian.

Proposition: local equilibrium sensitivity. Suppose the stated interior closure conditions hold, the vector field and observable $Q(z, \mu)$ are continuously differentiable locally, and $F_z(z^*, \mu)$ is nonsingular. Then a unique differentiable equilibrium branch exists in a neighbourhood of this root and parameter. Its derivatives are

$$\frac{dz^*}{d\mu} = -F_z^{-1} F_\mu, \quad (12)$$

$$\frac{dQ^*}{d\mu} = Q_\mu - Q_z F_z^{-1} F_\mu. \quad (13)$$

At a physical boundary the claim uses the smooth extension and is restricted to inward admissible perturbations. The result follows by differentiating $F(z^*(\mu), \mu) = 0$ and applying the implicit function theorem. It establishes local uniqueness of the branch, not global uniqueness or global existence. The dimensional dynamic Jacobian is $J = F_z/t_0$; this constant time scaling does not change (12). $\square$

The terms in (13) separate explicit parameter dependence from the response mediated by the complete state. For AE4 expression the outflow law has $Q_{e_4} = 0$. This mathematical zero does not mean that AE4 has no biological effect on secretion. Likewise, Q_z has only a lumen-volume

component on the active-outflow branch, but every retained network process can influence that volume.

The expression derivatives were checked against independently solved nearby equilibria at steps 10^{-4} and 5×10^{-5} , centred at $e_4 = 1, 0.5, 0.05$ and second-order inward at $e_4 = 0$. Validation covers all eleven coordinates and $Q, N, J_4, P, H, \text{pH}_i$. Across these checks, the largest observable component relative discrepancy is 1.034×10^{-7} ; the largest state-derivative relative discrepancy is 8.070×10^{-8} . The maximum nearby-root residual is 5.683×10^{-12} in the dimensionless independent equations. This validates local derivatives at the sampled points, without assigning those difference steps the meaning of uncertainty intervals.

8.4 Three distinct measures of NKCC compensation

Because N is an NKCC1 *cycle* flux carrying two chlorides, the local stationary chloride compensation gain is

$$C_N(e_4) = -\frac{2 \, dN^*/de_4}{dJ_4^*/de_4}, \quad dJ_4^*/de_4 \neq 0. \quad (14)$$

Here the parent J_4 carries one chloride per cycle. At a zero denominator the ratio is undefined and the two unnormalised derivatives must be given separately. The null entry below is an inward derivative and has a nonzero denominator even though $J_4^* = 0$ there.

e_4	dN^*/de_4	dJ_4^*/de_4	C_N
1	-0.0117368	0.0252674	0.929011
0.5	-0.0283978	0.0620609	0.915159
0.05	-0.0719960	0.160300	0.898266
0	-0.0795571	0.177417	0.896838

Flux derivatives in the table are in fmol/s per unit expression. Equation (14) describes incremental replacement at equilibrium; it does not quantify the complete knockout.

For the original finite-time experiment let $\mathcal{I} = [60, 600]$ s and superscripts W and 0 denote WT and AE4 null. The relative integrated NKCC increase and the integrated chloride replacement fraction are instead

$$R_N = \frac{\int_{\mathcal{I}} 2(N^0 - N^W) dt}{\int_{\mathcal{I}} 2N^W dt} = 0.2316345,$$

$$C_{N,\mathcal{I}} = \frac{\int_{\mathcal{I}} 2(N^0 - N^W) dt}{\int_{\mathcal{I}} (J_4^W - J_4^0) dt} = 0.8850704.$$

These signed integrals use the frozen one-second flux records, with nonzero denominators. NKCC supplies an additional 36.33701 fmol of chloride against 41.05550 fmol of missing AE4 chloride. Thus 23.16% is an increase relative to WT NKCC loading, whereas 88.51% is replacement of lost AE4 chloride over that specified window. Neither is the local gain C_N . The window also differs from the $[0, 600]$ s cumulative secretion comparison. The constant-stimulus stationary null secretion deficit is 0.94617%, compared with the frozen finite-time 3.8575%; their difference retains transient storage and state evolution.

8.5 Counting local cases and interpreting slow coordinates

The 26 equilibrium entries are class/expression *cases*: eight C0 expression values and three values for each of the other six source classes. At zero expression the AE4 source vanishes, so all seven null entries solve the same system. The count must not be read as 26 distinct equilibria at one parameter point. Twenty two entries pass the inherited physiological gates. Across all

cases the smallest singular value of the dimensional Jacobian is at least $9.202 \times 10^{-4} \text{ s}^{-1}$ and its condition number ranges from 1.882×10^4 to 2.931×10^4 in the declared scaled coordinates. Nonsingularity supports the local framework; the conditioning and numerical step checks qualify its implementation.

The slow eigenvectors involve volume and ionic storage together. For the slowest WT mode (345.82 s), intracellular potassium, cell volume and intracellular chloride account for 28.82%, 20.65% and 18.35% of the sum of absolute components of the scaled right eigenvector. For the slowest null mode (459.92 s), the corresponding shares are 31.04%, 24.68% and 16.55%. These are coordinate-dependent descriptive shares, not fractions of physiological energy, transported chloride or causal control. They supplement the secretion input/output residues and nonnormality diagnostics reported below. In particular, they give no justification for removing either volume or declaring the full chemical system instantaneous.

9 NBC removal: capacity obstruction versus secretion

At the parent WT root, NBC supplies $J_B/J_4 = 0.954815$ of the AE4 alkalinity demand. The conditional no NBC capacity bound covers 57.9613% of that particular demand. These are state and capacity comparisons; the total support ratio at equilibrium is identically one. Resolving the full system as NBC capacity decreases gives:

NBC capacity fraction	J_4^* (fmol/s)	J_B^* (fmol/s)	pH_i^*
1.00	0.075072	0.071680	7.0504
0.75	0.062485	0.058576	7.0331
0.50	0.047042	0.042486	7.0135
0.25	0.028351	0.022995	6.9918
0.00	0.006327	0.000000	6.9683

At zero NBC, a root satisfying the inherited physiological gates has $Q = 0.00153598288 \text{ pL/s}$ and $\text{pH}_i = 6.96833154$. The reference WT value is $0.00158406711 \text{ pL/s}$, so secretion falls by approximately 3.035%. AE4 loading falls from 0.07507245 to 0.00632651 fmol/s. The conditional obstruction is respected because the network selects a much lower AE4 demand. Additional alkalinity support is needed to sustain the larger stated loading level under the capacity assumptions; universal NBC necessity for all secretion is disproved for this frozen mathematical system by the valid counterexample root. Existence of this root does not establish a unique global attractor or identify the biological NBC isoform.

10 Independent equilibrium and spectral review

The full inherited physiological and conservation gates were applied to all 26 roots. Twenty two roots pass; the four excluded roots are retained in the table below. All 26 Jacobians are nonsingular and have negative real spectral parts on the 11 dimensional charge manifold. This establishes numerical local stability in the inward regulatory linearisation, not global uniqueness or global physiological validity.

Class	e_4	pH_i	$[\text{Na}]_i$	τ_{slow} (s)	Failed gate
C1	1	7.0139	45.694	564.4	na_i_mM
C2	1	6.5211	57.460	500.5	na_i_mM, ph_i
C4a	1	6.5214	31.108	455.7	ph_i
C4b	1	6.5792	15.581	313.6	ph_i

The first inherited checkpoint recorded 22 roots and four unsuccessful large continuation steps, without individual seeds, class labels or error strings. That aggregate failure record is preserved verbatim in the review JSON; missing details have not been reconstructed. The completed analysis follows C0 through eight expression levels. Other classes use the converged C0 WT as the WT seed and matching C0 roots for 5 percent expression and the null. This documented retry strategy obtains all 26 roots, including the four outside the physiological gates.

Eigenvalues were matched individually between relative coordinate difference steps 10^{-5} and 5×10^{-6} . The greatest matched relative eigenvalue discrepancy is 5.43e-08. At full regulatory activation the production evaluator clips values beyond one. A centred derivative across this boundary halves the regulatory Jacobian column. The corrected second order inward derivative gives the regulatory eigenvalue $-1/30 \text{ s}^{-1}$. The much slower chemical modes remain, with decay times 345.82 s in WT and 459.92 s in the null.

10.1 Thermodynamics at the local roots

Class	e_4	$A_4/(RT)$	Signed flux compatibility
C0	1	-0.9589	opposed
C1	1	3.8925	supported
C2	1	3.8589	supported
C3a	1	1.5135	supported
C3b	1	-3.7440	opposed
C3b	0.05	-2.2240	opposed
C4a	1	0.7093	supported
C4b	1	-4.3271	opposed
C4b	0.05	-2.3400	opposed

Positive forward affinity supports the proposed positive chloride cycle direction. The product of signed cycle flux and affinity separately checks the actual implemented flux. These diagnostics do not replace the inherited scalar cycle law. Every null has exactly zero AE4 flux, regardless of the hypothetical forward affinity. Thermodynamic opposition is retained and is not repaired by refitting.

10.2 Modal input and output projection

For $\dot{z} = Jz + B\delta e_4$ and $\delta Q = Cz$, let $J = V\Lambda V^{-1}$. The residue of mode k is $(CV)_k(V^{-1}B)_k$, and its stationary sensitivity contribution is this residue divided by $-\lambda_k$. The right vectors, inverse left rows and both projections are reported for every WT and null mode. The Euclidean norm uses the declared dimensionless coordinates and is not a physiological energy norm.

State	τ_{slow}	$\kappa(V)$	$\max_t \ e^{Jt}\ _2$	dQ^*/de_4
WT	345.82	22.02	1.37	1.28527e-06
Null	459.92	21.59	1.37	5.06256e-05

The propagator maxima are sampled numerical diagnostics over 0 to 10000 s, not proved suprema. State amplification, expression input projection and secretion output projection must be considered together. A small stationary secretion sensitivity can result from cancellation of modal contributions even when slow state modes are present. In WT, the 345.82 s and 223.48 s modes contribute -5.5754×10^{-5} and $+1.6013 \times 10^{-5} \text{ pL s}^{-1}$ per unit expression to the stationary secretion derivative. These contributions partially cancel faster positive contributions, leaving the much smaller net derivative $+1.2853 \times 10^{-6} \text{ pL s}^{-1}$. At the null, the 459.92 s mode contributes $-3.5795 \times 10^{-4} \text{ pL s}^{-1}$, against a net derivative of $+5.0626 \times 10^{-5} \text{ pL s}^{-1}$. The slow modes therefore do project

onto expression driven secretion and substantially shape its transient approach; their effects are not absent merely because the final WT derivative is small. For example, at 600 s the WT linear step derivative is $1.0027 \times 10^{-5} \text{ pL s}^{-1}$, still above its eventual value. The 30 s regulatory mode has zero expression input projection at both endpoints, because its target does not depend on expression. Its WT secretion output projection is nonzero, so a perturbation that actually excites regulation is a different experiment. There is therefore no basis for claiming that this model lacks a slow mode. The linear expression response at either endpoint is not a validated approximation to the complete WT to null change.

10.3 Independent reference and sensitivity verification

All 34 diagnostic columns at all 11 frozen display times were reproduced for WT, 5 percent expression and the null using the original Task 40 stimulus. The largest relative diagnostic discrepancy was 2.26e-07; the largest cumulative secretion discrepancy was 6.73e-10 pL. The original stimulus is unstimulated at the exact initial instant; the separate constant stimulus equilibrium experiment is fully stimulated there. This distinction matters for instantaneous transporter fluxes, even though initial outflow is state determined.

Implicit expression sensitivities for all independent coordinates and six observables were checked against separately solved nearby roots at 10^{-4} and 5×10^{-5} . Centred differences were used at expression 1, 0.5 and 0.05, and second order inward differences at the null. All component discrepancies and root residuals are provided in the review JSON. Chloride compensation retains the factor two for NKCC cycles.

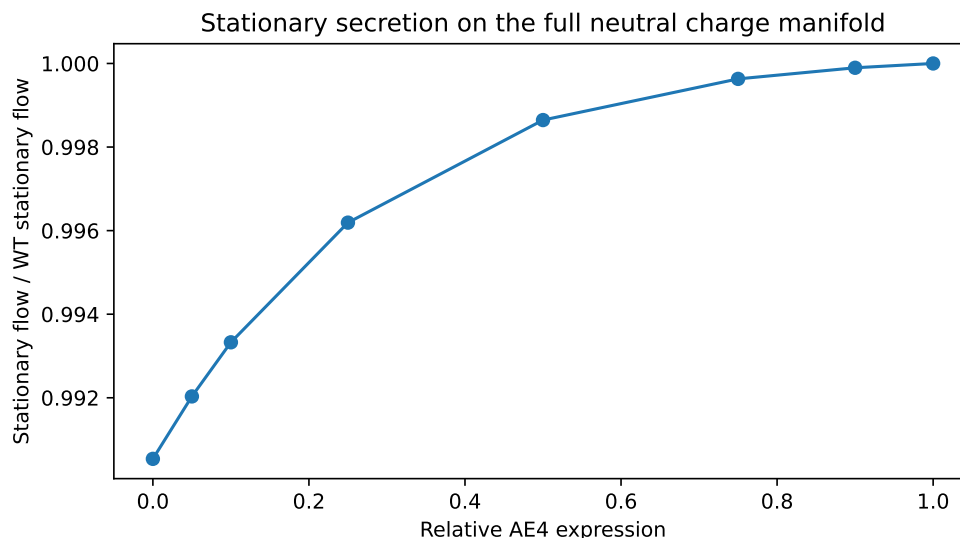


Figure 1: Constant stimulus C0 equilibrium secretion normalised by WT. The vertical axis deliberately resolves the small change around one; the complete null deficit is 0.94617%, not the 600 s cumulative deficit. Each point is a solved local root, not evidence of global uniqueness.

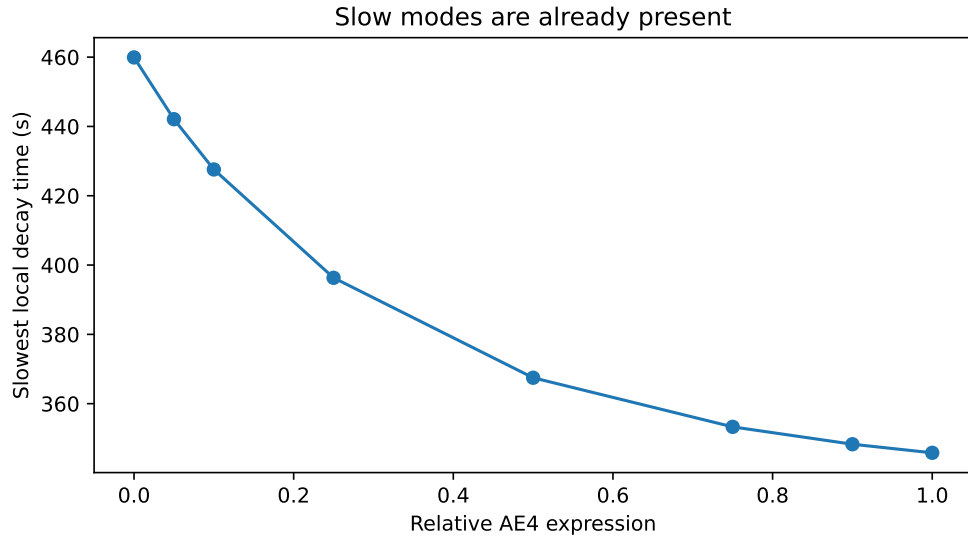


Figure 2: Dominant local decay times on the eleven dimensional neutral manifold during expression continuation. Carbon, alkalinity, both volumes, finite lumen composition and regulation are retained. These local timescales do not alone establish the shape of genotype divergence.

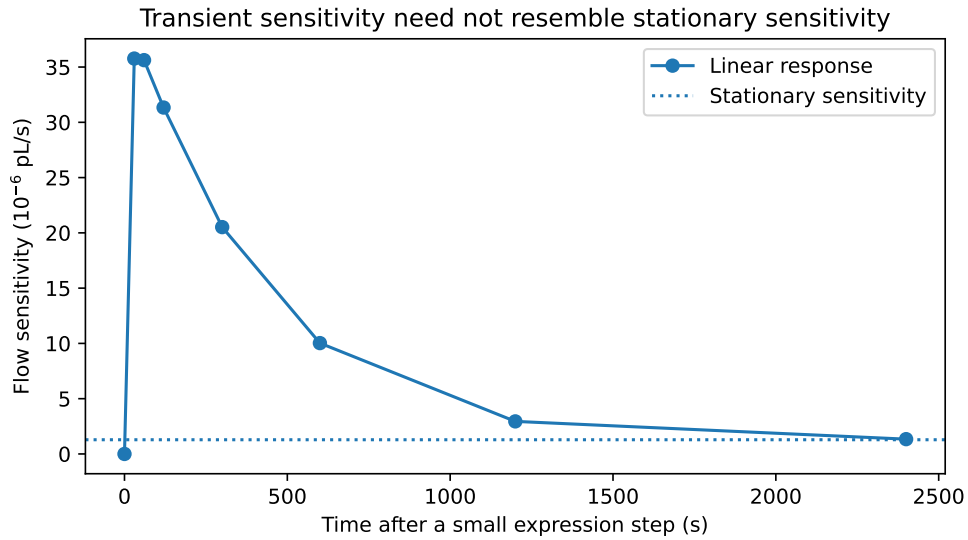


Figure 3: Local expression step response at the WT equilibrium. This linear response retains input and secretion output projection. It is a derivative experiment, not a replacement for a finite knockout trajectory from the inherited resting state.

11 All frozen source classes

The exact projections were derived above without class specific refitting. For completeness, the full expression root of every class is tabulated here. Sodium concentrations are in mM. Physiological acceptance and thermodynamic compatibility are separate tests; passing this table's gates does not erase the opposed fluxes reported in the spectral review.

Class	pH _i *	[Na] _i *	t _{slow} (s)	Failed criteria
C0	7.0504	18.083	345.82	None
C1	7.0139	45.694	564.38	sodium
C2	6.5211	57.460	500.52	sodium, pH
C3a	7.0220	24.508	431.19	None
C3b	7.0734	14.206	314.61	None
C4a	6.5214	31.108	455.72	pH
C4b	6.5792	15.581	313.64	pH

In particular C0 is a physiological mathematical root with thermodynamic opposition under the substituted ensemble source convention. Edge kinetics inherited from the carrier are insufficient to guarantee consistency of the whole substituted molecular cycle. The source vector analysis identifies the conserved signatures that require measurement; it does not validate the scalar kinetics for each molecular hypothesis.

12 The prescribed Task 41 inverse family

The Task 41 construction is target selected. Its added hypothesis changes stimulated CaCC conductance before the full electrical and NBC closure:

$$g_{\text{CaCC}}(e, a; \rho) = g_{\text{parent}}\{1 - a\rho(1 - e)\}, \quad \rho = 1 - b, \quad 0 \leq \rho < 1. \quad (15)$$

Here e is AE4 expression and a is the protocol normalised recruitment fraction used by NBC:

$$a = \min\left\{1, \max\left\{0, \frac{[\text{Ca}]_i - 0.058 \mu\text{M}}{(0.25 - 0.058) \mu\text{M}}\right\}\right\}.$$

The parent CaCC Hill gate remains a distinct retained factor. The full conserved system, carbonate closure, electrical closure, both volumes and AE4 regulation remain active. WT nests the parent exactly for every state and every ρ . At full activation, the null retains fraction $b = 1 - \rho$ of its parent CaCC conductance.

Every 600 s trajectory uses the identical frozen WT initial conserved state and original stimulus protocol. No genotype resting solve, new mechanism or parameter fit is introduced. All inherited positivity, concentration, pH, volume, source and conservation gates are checked at accepted solver endpoints and at the original observation times. The original WT stimulation requirement is also retained; it is not imposed retrospectively on mutants.

For $X \in \{A, S\}$ define

$$D_X(\rho, p) = 1 - \frac{I_{0,X}(\rho, p)}{I_{1,X}(p)}. \quad (16)$$

$I_{e,A}$ is the adaptive time integral of outflow and $I_{e,S}$ is the original trapezoidal observable on integer seconds, including $t = 0$ and 10^{-6} s. These are distinct numerical observations. The prescribed comparison threshold is $D_X = 0.303$: it is one reported standard error below the experimental mean of 35%, whose reported standard error is 4.7 percentage points. It is not a confidence limit, a measured biological range or an admissibility gate.

12.1 Reproduction and numerical crossing

The selected Task 41 parameter is $b = 0.10511872843288446$, or $\rho = 0.8948812715671155$. The three selected trajectories reproduce the frozen sampled cumulative outputs to within $6.7\text{e-}10$ pL.

Case	Sampled output (pL)	Adaptive output (pL)	Sampled deficit (%)
WT	0.992524545	0.992544443	0
5% AE4	0.762622459	0.762624862	23.163365
AE4 null	0.692175197	0.692173709	30.261151

In particular, the frozen selected null deficit is slightly below 30.3%. It must not be described as exactly attaining this new comparison threshold.

An ordered scan of 17 recruitment fractions, followed by bracketed numerical root refinement in the prescribed family, gives:

Observable	Crossing ρ	AE4 dependent recruitment (%)	Residual b
Adaptive	0.8950726882	89.5072688	0.1049273118
Original sampled	0.8950806285	89.5080628	0.1049193715

Both crossings pass every inherited 600 s gate. At the adaptive crossing, the original sampled deficit is 30.2984507%; at the sampled crossing, the adaptive deficit is 30.3015492%. Refinement at relative solver tolerances 10^{-8} and 2×10^{-9} , with respective maximum steps 2 s and 1 s, changes the two crossing fractions by $1.4\text{e-}13$ and $1.7\text{e-}11$, respectively. Independent integration of the 13 production states, followed by adaptive quadrature of dense flow, also reproduces the selected cases and crossing.

The deficit increases at the sampled points, but monotonicity between them and exclusion of all earlier unsampled crossings have not been proved. These are the first numerical crossings found in this prescribed family, not global lower bounds, universal recruitment requirements or independent validation of the coupling hypothesis.

12.2 The separate extension beyond 600 s

Continuing constant full stimulation to 3600 s gives the following null diagnostics. Each run passes all inherited gates through 600 s. Intracellular pH is the only later gate failure; all other monitored gates continue to pass.

Case	First pH = 7.3 time (s)	pH at 3600 s	Limiting local root pH
Selected Task 41	1486.30957	7.3117946	7.3120147
Adaptive crossing	1484.74517	7.3118437	7.3120644

The limiting roots found from these late states are mathematical continuations outside the inherited pH domain. They do not establish globally unique equilibria or physiological long term validity. These separate diagnostics do not change any frozen 600 s Task 41 result.

12.3 Local uncertainty of the crossing

For a locally regular numerical crossing with $\partial_\rho D_X \neq 0$,

$$\frac{d\rho_X}{d \log p} = -\frac{\partial_{\log p} D_X}{\partial_\rho D_X}, \quad \frac{d \log b_X}{d \log p} = -\frac{1}{1 - \rho_X} \frac{d\rho_X}{d \log p}. \quad (17)$$

Eight effective or architecture derived quantities were selected using the Task 43 provenance classification. A perturbed WT denominator is used with the same perturbed parameter in the

null. The conserved initial state is always held at the inherited WT value. Thus changing the buffer amount also changes the algebraic initial pH; it does not trigger a new resting solve.

The following elasticities refer to the adaptive crossing. The companion sampled results are retained in the machine readable crosswalk.

Parameter	$d \log \rho / d \log p$	$d \log b / d \log p$
NBC capacity	-0.0059550	+0.0507982
AE4 carrier amount	-0.0050569	+0.0431371
NKCC effective scale	-0.0021926	+0.0187041
NHE1 carrier amount	+0.0008018	-0.0068396
Pump capacity	-0.0355825	+0.3035333
CaCC conductance	+0.0983320	-0.8388120
Cell buffer amount	+0.0071557	-0.0610411
Apical water permeability	-0.0002975	+0.0025380

The derivative calculation is checked against independently solved nearby crossings at logarithmic perturbation sizes 10^{-3} and 5×10^{-4} . Every perturbed WT and null trajectory passes its inherited gates. The maximum relative discrepancy between implicit crossing derivatives and nearby root differences is $1.7\text{e-}06$. The larger relative sensitivity of residual recruitment b is relevant when interpreting a conductance assay. None of these eight quantities has an established experimental uncertainty interval in the audited configuration. Accordingly, these calculations establish local sensitivity only; they do not prove robustness over a finite range or identify a finite parameter change that overturns the conclusion.

12.4 A discriminating experiment and scope

At the numerical comparison crossing the null must retain approximately 10.49% of stimulated CaCC conductance in this family. A direct test would compare stimulated CaCC conductance and channel recruitment in WT and AE4 loss cells while matching calcium activation, voltage, intracellular chloride and pH, so that a changed driving force is distinguished from a changed conductance. Surface channel abundance and channel open probability would help distinguish recruitment from gating. Concurrent measurements of pH, chloride, cell volume and secretion over the first ten minutes and after approximately 25 minutes would test the predicted coupled response and its later pH failure. A secretion measurement alone does not identify this specific dependency.

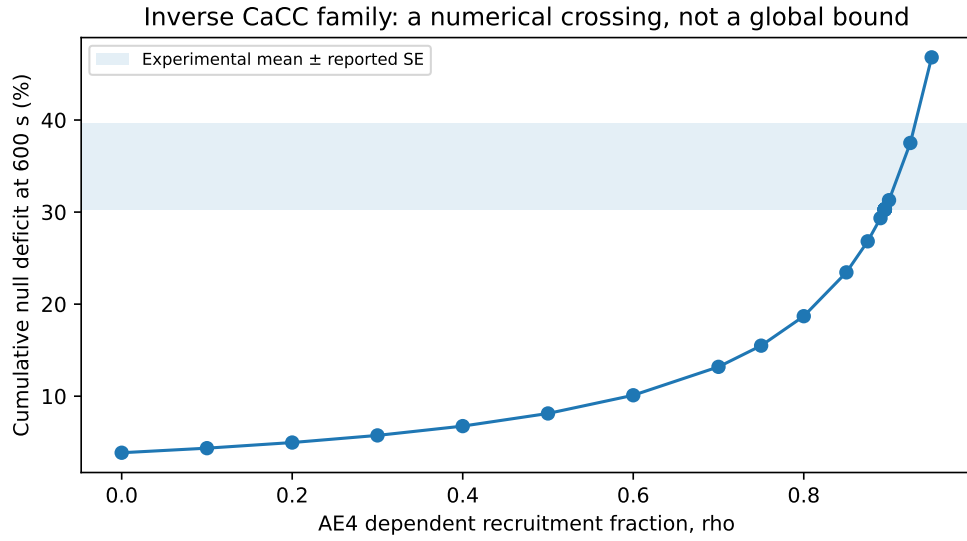


Figure 4: Ordered numerical scan within the one prescribed Task 41 family. The experimental band represents the reported mean plus or minus one standard error, not a confidence interval. The refined adaptive and sampled crossings are recorded separately. Neither a global lower bound nor global monotonicity follows from the plotted points.

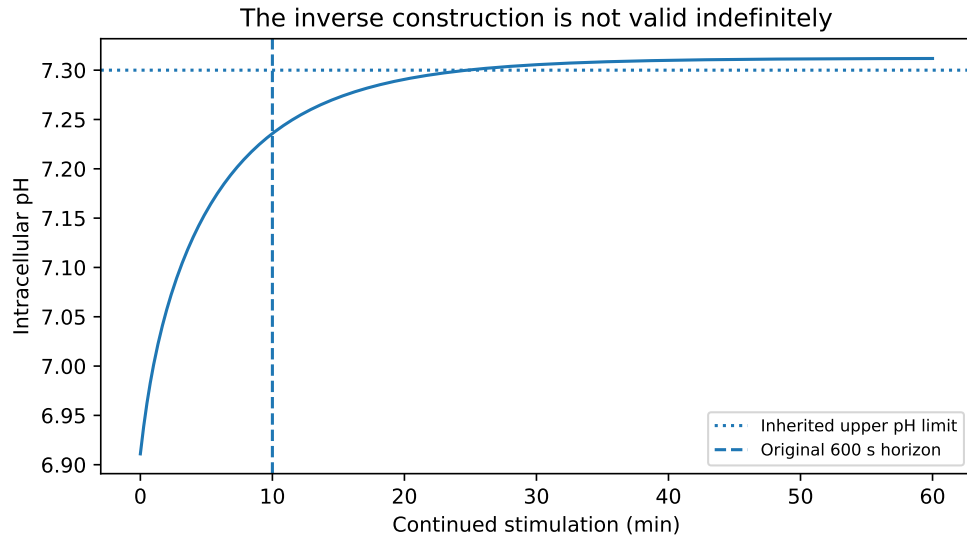


Figure 5: Separate extension of the adaptive crossing null beyond the declared 600 s Task 41 window. It first crosses the inherited upper pH gate near 1485 s; the selected case has a nearly identical late crossing. The later diagnostic does not revise the frozen 600 s observable; it limits claims of sustained physiological validity.

13 Dependence on uncertain parameters

13.1 Sensitivity crosswalk with Task 43

The sixteen selected parameters are uncertain effective settings or inherited calibrated capacities. Each is perturbed independently by $p \exp(\pm h)$, with $h = 10^{-3}$ and 5×10^{-4} ; both genotypes are solved afresh at each perturbed point. These steps are numerical checks and do not define uncertainty intervals. No defensible physiological interval is available from the provenance audit, so no global robustness claim is made.

Parameter	E_Q	$\partial \text{pH} / \partial \log p$	E_D	E_{C_N}	$E_{\tau_{WT}}$
NBC capacity	0.01992	0.06483	2.085	-0.06438	0.107
AE4 amount	0.0008114	-0.07246	0.08494	0.02339	-0.06397
NKCC scale	0.05536	-0.001558	-1.494	0.06404	0.1901
NHE amount	-0.003261	0.07084	-0.2293	0.007679	-0.01473
Pump capacity	0.5259	0.1206	4.634	-0.2734	-1.522
CaCC conductance	0.03247	-0.005524	-0.2212	0.01105	0.008274
Cell buffer amount	-0.001575	0.001396	-0.2063	0.009504	0.3086
Apical water permeability	0.002129	-8.317e-05	-0.003558	0.0002239	-3.74e-05
AE4 activation time	0	0	0	0	0
AE4 activation gain	0.0001623	-0.01449	0.01699	0.004678	-0.01279
Apical pump fraction	0.04956	-0.003571	-0.05983	0.003151	0.04517
NHE stimulation gain	-0.003261	0.07084	-0.2293	0.007678	-0.01473
Basolateral CO ₂ permeability	-0.004448	-0.08319	-0.4338	0.05062	-0.06818
AE2 capacity	-7.335e-05	0.000363	-0.03056	-0.03383	7.508e-05
NBC saturation width	-0.01435	-0.0467	-1.502	0.01777	-0.07263
NKCC stimulation gain	0.05536	-0.001558	-1.494	0.06404	0.1901

Here $E_Y = \partial \log Y / \partial \log p$ and $D = 1 - Q_{\text{null}} / Q_{\text{WT}}$ is the stationary deficit. Its small baseline value makes E_D potentially large; the machine readable table also supplies percentage point derivatives. pH is already logarithmic, so its unnormalised derivative is shown; the hydrogen concentration elasticity is $-\log(10) \partial \text{pH} / \partial \log p$. All derivatives are partial sensitivities at the inherited parameter point, without recalibration. Pump capacity has the largest local effect on WT secretion and on the stationary null deficit. The latter is also sensitive to NBC capacity and its assumed saturation width, and to effective NKCC capacity. These are priorities for experimental constraint. The calculations do not show that a finite parameter change overturns a conclusion, and cannot establish that the small null phenotype persists throughout an unknown uncertainty range. The exact balance identities survive these capacity changes within the specified architecture; their numerical consequences need not. The equilibrium total alkalinity support ratio is identically one and cannot demonstrate robustness. The saved results instead give the NBC share J_B / J_4 , the observed non NBC share $(J_H - J_2) / (2J_4)$, and a conditional capacity ratio $(H_{\text{max}} + J_{2,\text{max}}) / (2J_4)$. The bound assumes the fixed bath, positive sodium and $\text{pH}_i \geq 6.6$. It permits AE2 to supply alkalinity at its maximal reverse capacity. Its local sensitivity is useful, but it does not prove the available capacity over an unknown parameter interval. The AE4 activation time has no local effect on constant stimulus equilibria. Its eigenvalue is $-1/\tau_4$ and is separated from the slower chemical mode at this point; a zero local slow mode sensitivity does not establish that activation timing is irrelevant to finite time secretion. The buffer calculation holds the fixed charge and impermeant osmole pools fixed while changing only buffer amount. NKCC capacity and its stimulated multiplier produce the same stationary perturbation because only their product enters at constant full stimulation. Independent root perturbations validate the IFT derivatives at both steps, and expression perturbations validate the chloride compensation gain at both steps for every nearby parameter point. Fixed state parameter derivatives use the smaller steps 10^{-5} and 5×10^{-6} to resolve cancellation in the buffer response. Decay sensitivities additionally use fourth order differentiation of the chemical block at two steps, together with the exact regulatory eigenvalue; the block triangular structure retains all eleven dynamical eigenvalues.

The supplied files retain root residuals, physiological gate results, Jacobian step checks and all derivative comparisons. The separate inverse family calculation supplies the Task 41 crossing sensitivity. The independent complete gate supplement resolves all 128 perturbed WT and null parameter roots and 256 additional expression roots used to check compensation. Every numerical solve succeeds and every inherited gate passes. Its largest conservation residual divided by its own tolerance is 4.086×10^{-5} ; the largest independent RHS residual is 6.880×10^{-12} . This supplements the original derivative checks without weakening a gate. Exact reproduction of this supplementary calculation from the clean numerical checkout is recorded separately.

14 Comparison with experimental evidence

The gland phenotype and the comparison threshold. In the 2015 mouse submandibular gland study, stimulation used $0.3 \mu\text{M}$ carbachol and $5 \mu\text{M}$ isoproterenol. AE4 deletion reduced total secretion over ten minutes by $35 \pm 4.7\%$ (mean $\pm$ standard error; six glands per genotype). Initial flow over two to three minutes was comparable, with the deficit developing later. AE2 deletion had no detectable flow effect against its own control strain. AE4 deficient acinar cells had lower resting chloride and slower chloride reuptake; initial chloride exit and resting pH were not detectably changed. These are distinct measurements, rather than interchangeable manifestations of a single steady flux [4]. The prescribed $30.3\% = 35\% - 4.7\%$ inverse target is therefore a comparison value one reported standard error below the mean. It is not a confidence bound, an uncertainty interval for a model parameter, or a universal biological minimum.

The Task 40 cumulative null deficit of 3.8575% establishes a small effect for the frozen model and protocol. The equal routing identity does not establish that small magnitude for every admissible parameter choice. Nor does agreement at 600 s establish the observed time course. The present simulations initialise every genotype at the inherited WT resting state, whereas the experiment used established knockout animals. That difference should remain explicit when comparing the initial response or stored intracellular chloride.

Compensation and pH. The 2015 isolated NKCC1 assay used bicarbonate free solution, $30 \mu\text{M}$ ethoxyzolamide and $50 \mu\text{M}$ T16Ainh-A01; bumetanide identified the NKCC1 dependent component. Initial chloride uptake showed no detectable genotype difference. Stimulation induced NHE dependent alkalisation also showed no genotype difference [4]. Neither finding measures the complete stimulated network integral. Consequently, the approximately 23.16% rise in integrated NKCC1 loading in the frozen Task 40 null is an unresolved experimental tension. It must not be described as observed compensation. The local equilibrium compensation derivative, the relative integrated NKCC1 increase and the fraction of missing AE4 chloride replaced by NKCC1 are three separate quantities throughout this report. Unchanged assay alkalisation does not establish equal NHE1 net flux at different ionic states, and unchanged resting pH does not identify individual alkalinity sources. A discriminating experiment would measure chloride loading, pH, sodium and volume through the same combined stimulus, with pathway specific perturbations and parallel controls.

What the 2018 model actually established. The published model predicted an approximately 24% flow reduction after AE4 deletion. Its reduced flow settled within about two minutes; the authors explicitly contrasted this with experimental development over roughly ten minutes. It predicted little NKCC1 activation following either exchanger deletion. It included carbon dioxide hydration, NHE1, AE2 and AE4, without another proton or bicarbonate transporter, and held lumen volume constant [6]. Those are historical model properties, not experimental determinations of the full network. The present retained carbon, alkalinity and volume coordinates make the storage terms explicit. Numerical slow modes and their projection onto secretion must be examined

before attributing a temporal discrepancy to absent slow regulation. A slow eigenvalue alone does not reproduce delayed genotype divergence.

Regulation and the inverse CaCC family. The 2021 study found that beta adrenergic stimulation increased native AE4 activity, that H89 prevented the response, and that constitutively active PKA increased AE4 activity in CHO cells. The S173A mutant lost PKA activation, whereas S273A retained it. The authors described phosphorylation of S173 as a probable mechanism; H89 is a nonselective inhibitor [5]. This supports regulated AE4 activity, but does not measure the model's AE4 regulatory gain, relaxation time or an AE4 to CaCC recruitment law. The Task 41 inverse crossing quantifies a requirement within one family selected using the phenotype. It supplies no independent validation of that family. Its direct experimental test is the fraction of stimulated apical chloride conductance lost upon AE4 perturbation at matched calcium, membrane potential, chloride, pH and volume, resolved from the initial response into sustained secretion. This distinguishes altered channel recruitment from changed driving force and from intracellular chloride storage.

Molecular source classes. Catalán and colleagues studied human AE4 expressed in HEK293 cells, combining mutagenesis, transport assays and molecular dynamics. The T756A–T448I mutant lost transport with extracellular sodium while retaining activity with potassium. Their discussion proposes both bicarbonate and carbonate cycles, including exchanged sodium and potassium, and explicitly calls for further experiments to resolve the alternatives [1]. Thus the seven frozen classes are specified source hypotheses, not seven experimentally established full cycles. Their conserved projections and equilibrium tests separate chloride, carbon, alkalinity and charge consequences under inherited scalar kinetics. Matching one projected signature cannot establish identical molecular mechanisms or identical whole cell responses. An unsuccessful root search cannot establish nonexistence. A thermodynamically opposed or nonphysiological case must retain that qualification. Joint measurements of cation and anion flux ratios, reversal conditions and current would constrain the source vector; rates measured across those gradients would constrain the separate kinetic law.

The most urgent missing measurements. The evidence checked here does not establish the identity, abundance or capacity of the assumed inward $1\text{Na} : 2\text{HCO}_3$ pathway. The alkalinity proposition gives a conditional requirement for additional support when the admissible NHE1 and AE2 contribution is insufficient. It does not prove that a particular NBC isoform is universally necessary. The assumed loading allocation used to construct NBC capacity is not a measured transporter partition. Measurements of sodium and bicarbonate influx with current and pH during combined stimulation are therefore a priority, alongside native NHE1 and NKCC1 capacity and state dependence. The sensitivity crosswalk identifies which effective capacities, conductances and buffer settings matter most at the analysed point. Until independent ranges constrain them, local elasticities do not establish robustness over biological uncertainty.

Scope of source verification. The comparison uses the 2015 primary full text and Figures 1 and 2, the repository copy of the 2018 published paper, the 2021 primary abstract and figure captions, and the 2025 primary author manuscript including its discussion. These four references were checked to that extent. No systematic literature survey, new experimental measurement or complete source audit of every inherited kinetic coefficient is claimed here; parameter provenance is recorded separately in Task 43.

15 What is established and what remains unresolved

Status	Conclusion and scope
Proved or exact	Unique mathematical carbonate and electrical roots under the stated positivity and slope assumptions, smooth dependence inside strict executable brackets, local well posedness, exact retained nondimensionalisation, alkalinity storage budget, conditional capacity obstruction and equal routing chloride identity. Joint source projections have rank five.
Local numerical	Twenty six class/expression cases yield nonsingular stable local roots; twenty two pass the physiological gates. WT/null slow times are 345.82/459.92 s. The stationary null deficit is 0.94617%; the WT local chloride compensation is 92.901%. Local IFT and parameter derivatives are independently verified. A physiological NBC absent equilibrium exists.
Finite time numerical	Frozen Task 40 observables reproduce. The 60–600 s integrated NKCC increase is 23.163% and missing AE4 chloride compensation 88.507%. First numerical Task 41 crossings need about 89.51% AE4 dependent stimulated CaCC recruitment. The 600 s gates pass; selected continued null cases later violate the pH gate.
Unresolved	Global uniqueness, stability and positivity; a justified reduced model; global exclusion of earlier inverse crossings; robustness over unknown parameter ranges; native carrier abundance and pathway identity; molecular source stoichiometry and compatible kinetics; the isolated NKCC assay tension; independent validation of the Task 41 CaCC coupling.

Several earlier interpretations require explicit correction. Stationary cancellation does not imply zero total sensitivity or a uniformly small phenotype. The system already contains slow chemical modes; the corrected regulatory derivative gives 30 s, not the spurious 60 s obtained by crossing the production clamp. The initial unsuccessful continuation record is not evidence of nonexistence. NBC is not universally necessary for secretion in this frozen system, though it supplies most of the larger AE4 alkalinity demand. The old selected Task 41 point gives a sampled deficit of 30.26115%, slightly below 30.3%, and the refined crossing depends on the declared integral. The comparison value is mean minus one standard error. The construction remains target selected and its late pH failure must accompany any claim about sustained stimulation.

The immediate experimental priorities are functional pump capacity and apical distribution; NBC equivalent sodium/bicarbonate flux, current and its saturation curve; ion conditioned NKCC and NHE capacity; simultaneous TIC, buffering, pH and volume measurements; and stimulated CaCC conductance and recruitment under acute AE4 perturbation at matched calcium and driving forces. Source stoichiometry requires combined cation, chloride, carbon and current measurements. These measurements would constrain the effective parameters that dominate local sensitivities and separate a transport pathway from a fitted phenomenological coupling. An accepted physiological point and small numerical perturbations do not supply those missing ranges.

16 Reproduction, verification and deliverables

All new material is under `analysis/44_full_system_mathematics/`. The frozen scientific base is `c4d4f207f702d8eee09ab94d2d90ae90552dd641`. The original source manifest, active parameter objects, source hashes and recovery record distinguish scientific files from historical metadata changes. The complete numerical checkpoint was cloned independently, checked out detached, and regenerated with empty inverse trial caches before this standalone report was written. All eighteen

commands passed. Forty six regenerated authoritative JSON/CSV files plus one unchanged historical record were compared: all 331157 numeric values matched bit for bit. Only elapsed runtime and an ancestor validated verification commit field were excluded. All 532 inverse trial files were regenerated; the optional cache is not part of the published scientific deliverables. All 1751 tracked files outside Task 44 were unchanged in that clean replay.

File or group	Content
README.md, RECOVERY.md	Reproduction commands, recovered history and derivative correction.
core.py, run_analysis.py	Exact coordinates, production adapter, scaling, continuation, local derivatives and baseline reproduction.
verify_exact.py	Independent full RHS reconstruction, pH and electrical closure checks, exact source and balance identities at valid states.
equilibrium_review.py	Full physiological gates, matched eigenvalues, modal projection and independent nearby roots.
sensitivity_analysis.py	Sixteen uncertain parameter sensitivities and derivative verification; Task 43 crosswalk source.
verify_sensitivity_gates.py	All 384 nearby roots tested against the complete inherited gates, with solver failures retained separately.
inverse_analysis.py, inverse_verify.py	Original family, both crossing observables, long diagnostics, eight parameter sensitivities and independent thirteen state integration/quadrature.
output/, inverse_detail/	Machine readable source audit, scales, trajectories, roots, spectra, derivatives, projections, crossing scan, refinement, gates and verification.
output/clean_recomputation_verification.json	Exact numerical checkpoint, commands, file comparisons, exclusions and protected file audit.
report.tex, build.sh, sources.bib	Standalone report source, bibliography and clean build command.
FINAL_STATUS.md	Final verified checkpoint, findings, failures, caveats, artifact paths and checksums.

The PDF build resolves every citation and reference. Figures are generated from saved verified results; axis ranges and the distinct observables are stated in captions. The report, TeX fragments, vector figures and complete machine readable outputs are retained for scientific review. No model source, frozen output, Task 1–42 analysis or manuscript text was changed. The two analysis branches remain separate and unmerged.

References

- [1] Marcelo A. Catalán, Lisandra Flores-Aldama, Fernanda Fernández, Daniel Bustos, Natalia Apablaza, Ailen Hidalgo, Yuliet Mazola, Ella Matamala, Li Yo Kao, Ira Kurtz, Carlos Spichiger, José M. Sarmiento, Sebastián Brauchi, Wendy González, Leandro Zúñiga, and Gaspar Peña Münzenmayer. Molecular determinants of bicarbonate and cation transport in the human

- cation-dependent chloride–bicarbonate exchanger AE4. *American Journal of Physiology-Cell Physiology*, 328(6):C2070–C2084, 2025.
- [2] Chae Young Cha, Chiaki Oka, Yung E. Earm, Shigeo Wakabayashi, and Akinori Noma. A model of sodium–proton exchanger and its central role in regulation of ph and sodium in cardiac myocytes. *Biophysical Journal*, 97(10):2674–2683, 2009.
 - [3] Laurence Palk, James Sneyd, Trevor J. Shuttleworth, David I. Yule, and Edmund J. Crampin. A dynamic model of saliva secretion. *Journal of Theoretical Biology*, 266(4):625–640, 2010.
 - [4] Gaspar Peña Münzenmayer, Marcelo A. Catalán, Yusuke Kondo, Yasna Jaramillo, Frances Liu, Gary E. Shull, and James E. Melvin. Ae4 (Slc4a9) anion exchanger drives chloride-uptake-dependent fluid secretion by mouse submandibular gland acinar cells. *Journal of Biological Chemistry*, 290(17):10677–10688, 2015.
 - [5] Gaspar Peña Münzenmayer, Yusuke Kondo, Constanza Salinas, José Sarmiento, Sebastián Brauchi, and Marcelo A. Catalán. Activation of the Ae4 (Slc4a9) cation-driven chloride–bicarbonate exchanger by the camp-dependent protein kinase in salivary gland acinar cells. *American Journal of Physiology-Gastrointestinal and Liver Physiology*, 321(6):G628–G638, 2021.
 - [6] Elías Vera-Sigüenza, Marcelo A. Catalán, Gaspar Peña Münzenmayer, James E. Melvin, and James Sneyd. A mathematical model supports a key role for Ae4 (Slc4a9) in salivary gland secretion. *Bulletin of Mathematical Biology*, 80:255–282, 2018.